Kazakh-British Technical University

Data collection and preparation

Students' Independent Study (SIS) №1

Highest-Grossing Films: What Makes a Box-Office Hit?

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1. Objective

Every year a few films earn more than one billion dollars. We asked what these films have in common: (1) does a bigger budget mean a bigger box office, (2) do highly rated films earn more, (3) which genres make the most money, and (4) how box office changed over the decades.

2. Data collection

Web scraping. We scraped the Wikipedia page "List of highest-grossing films" with requests and BeautifulSoup after checking robots.txt (access allowed). The page has 88 tables, so our parser finds the right ones by their column names, not by position, and expands rowspan/colspan cells into a full grid. We took two tables: the 50 highest-grossing films of all time (rank, title, worldwide gross, year) and the highest-grossing film of each year since 1915 (153 rows, with budget).

API. For every film we sent two requests to the TMDb API: /search/movie (title + year) to get the TMDb id, and /movie/{id} to get budget, revenue, genres, rating, number of votes and runtime. The best search result was chosen by title similarity (difflib) plus a bonus for the same year. The API key was read from a local file excluded by .gitignore, so it is not in the public repository. All 183 films were found.

3. Cleaning and merging

- **Money text:** Wikipedia cells such as "T $2,257,906,828" or ranges like "$128,000,000–145,000,000" were parsed with a regular expression (first amount; the middle value for a range).
- **Titles and types:** footnotes [1] and the "†" mark (film still in cinemas, saved as a separate column) were removed; rank, year and dates were converted to numbers and datetime.
- **Duplicates:** 20 films appear in both Wikipedia tables (e.g. Avatar), so 203 rows became 183 unique films (key = normalised title + year).
- **Hidden missing values:** TMDb writes 0 when the budget (20 films) or revenue (16 films) is unknown. We changed 0 to NaN and used the Wikipedia budget for those 20 films. We did not fill values with averages, to avoid distorting correlations.
- **Quality check:** Wikipedia gross and TMDb revenue have a median difference of only 0.4% (167 films with both values), which confirms that films were matched correctly.

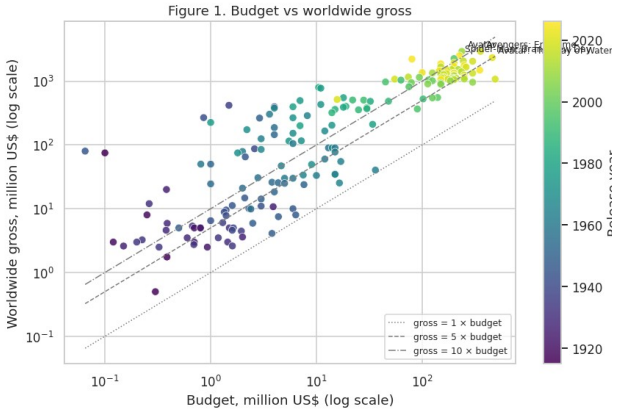


Figure 1. Budget vs worldwide gross (log scales)

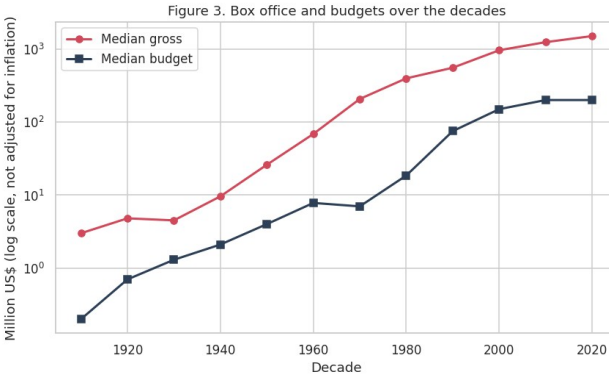


Figure 3. Median gross and budget by decade

4. Results

- **Budget matters, but less for modern films.** Across all 183 films, budget and gross are very strongly related (Spearman ρ = 0.89; log-log model R^2 = 0.75). For films released since 2000 the relationship is only moderate (ρ = 0.36, n = 58): when almost every blockbuster costs $150–200 million, the budget no longer separates the winners.
- **Ratings are not the key.** Rating and gross show a moderate correlation overall (ρ = 0.37), but for films since 2000 it falls to ρ = 0.12. Part of the overall link comes from time: newer films have much higher gross (ρ = 0.93 with year) and also slightly higher ratings (ρ = 0.37 with year).
- **Blockbuster genres.** Adventure is the most common genre (47% of films). Science Fiction (median gross $1,126 M), Action ($1,066 M) and Adventure ($1,026 M) earn the most, while Drama ($24.5 M) and Romance ($18.4 M) dominate only the early decades.
- **Growth over time.** The median gross of the top films rose from $3.0 M in the 1910s to $1,503 M in the 2020s, and the median budget from $0.2 M to $200 M (nominal dollars).
- **Best return on budget.** Old low-budget hits lead: Mom and Dad (1945) earned 1,231 times its budget, Snow White and the Seven Dwarfs (1937) 281 times and Rocky (1976) 225 times. Modern record holders return 5–12 times (Avengers: Endgame 8.2×, Avatar 12.3×).

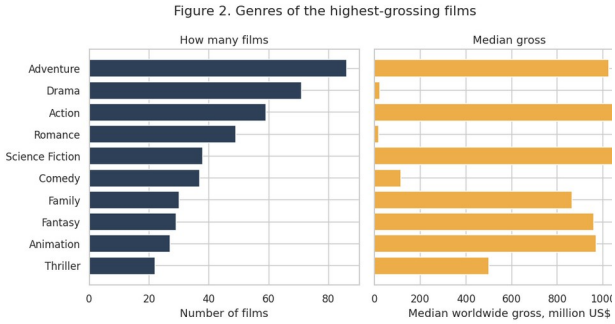


Figure 2. Genres: number of films and median gross

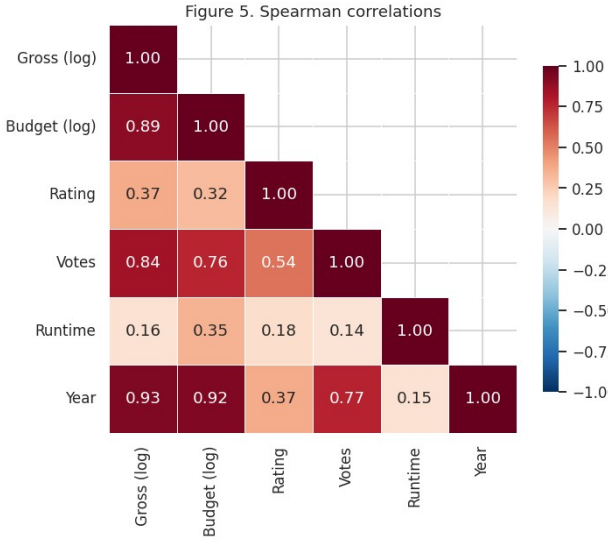


Figure 5. Spearman correlation matrix

Figure numbers follow the notebook, which contains all six figures and the full code.

5. Key insights

For the biggest hits, a large budget is close to a requirement rather than a guarantee: it explains the difference between old and modern films, but much less the difference between modern blockbusters. Franchise-friendly genres (adventure, action, science fiction, animation) are what the top of the list is made of, while reviews play a small role. Comparing two independent sources was useful: their close agreement (0.4%) shows the data can be trusted, and the few large gaps revealed errors in TMDb.

6. Limitations and conclusion

Money is in nominal dollars, not adjusted for inflation. The data contains only successful films, so it cannot explain why films fail (survivorship bias). TMDb budgets are partly estimates, and films marked † are still earning. Overall, we built a complete pipeline — scraping, API collection, cleaning, matching, merging, analysis and visualisation — and produced one clean dataset of 183 films with values from both sources.